Database II Lab/ 3rd Grade

[Second Semester] and [2026]

[Lab 5] [3/3/ 2026] [8.30 + 10.30 am]

Instructor Information

Instructor	Email	Hours
Dr. Rasool Hisham Dr. Zainab Namh Dr. Azhar Flaih Assist. Prof. Zahraa Jaaz	[rasool.hisham@nahrainuniv.edu.iq]	[2 Hrs]

Manipulating a MySQL Database in C# Windows Forms Application

Lab Objective:

- The objective of this lab is to develop a C# Windows Forms application that connects to a MySQL database and performs essential data manipulation operations (INSERT, UPDATE, DELETE).
- The lab also aims to demonstrate how to safely execute SQL commands using parameterized queries.
- Handle execution results and user notifications within the application.

In this lab, a connection is established with **company** MySQL database, and then three main operations are performed on the **department** table.

ExecuteNonQuery() Method

In C# with MySQL, **ExecuteNonQuery()** is a method of the **MySqlCommand** class that is used to execute **SQL statements that do not return data**. It is specifically used for:

- **DML Statements:** INSERT, UPDATE, and DELETE.
- **DDL Statements:** CREATE TABLE or ALTER TABLE.
- **Return Value:** It returns an **integer** representing the number of rows modified. If no rows were changed, it returns 0.

```
int rowsAffected = cmd.ExecuteNonQuery();
```

Steps to Insert a Record into MySQL Database:

1. **Define the Connection String**
   Specify the database connection details such as server name, username, password, and database name.

```
string connectionString = "server=...;user=...;password=...;database=...;";
```

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2. **Create a Connection Object**
   Initialize a connection to the MySQL database using MySqlConnection.

```
MySqlConnection conn = new MySqlConnection(connectionString);
```

3. **Open the Connection**
   Establish the connection to the database.

```
conn.Open();
```

4. **Define the SQL Query**
   Write an INSERT statement to add a new record into the department table.

```
string sql = "INSERT INTO department(dno, dname) VALUES (6, 'Test department')";
```

5. **Create a Command Object**
   Use MySqlCommand to prepare the SQL query for execution.

```
MySqlCommand cmd = new MySqlCommand(sql, conn);
```

6. **Execute the Query**
   Run the command using ExecuteNonQuery() to insert the record into the database.

```
cmd.ExecuteNonQuery();
```

7. **Handle Exceptions**
   Use a try-catch block to handle any errors that may occur.

```
try
{
   cmd.ExecuteNonQuery();
}
catch (Exception ex)
{
   MessageBox.Show(ex.Message);
}
```

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8. **Close the Connection**
   Always close the database connection after completing the operation.

```
conn.Close();
```

Note that, *If the department ID already exists, you get a primary key violation error!*

User Interaction: MessageBox

Is a method in C# used to display a message box in Windows Forms applications. It is part of the System.Windows.Forms namespace and is commonly used for showing alerts, confirmations, or information to users.

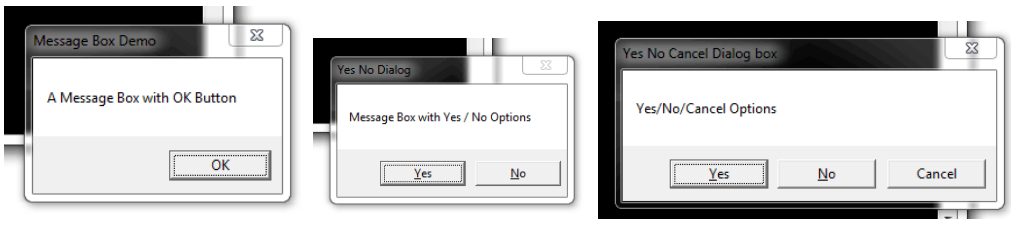
```
MessageBox.Show("message", "title", MessageBoxButtons.buttons, MessageBoxIcon
.icon);
```

Parameters:

- **Message**: The message to be displayed.
- **Title**: The title of the message box (optional).
- **Buttons**: The set of buttons to display.
- **Icon**: The icon to display on the message box.

You can specify different types of buttons that will appear in the message box using the **MessageBoxButtons** enumeration. Here are the most common options:

- **MessageBoxButtons.OK**: Displays only an "OK" button.
- **MessageBoxButtons.OKCancel**: Displays "OK" and "Cancel" buttons.
- **MessageBoxButtons.YesNo**: Displays "Yes" and "No" buttons.



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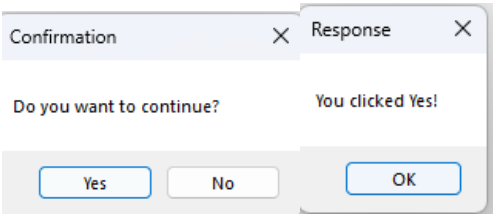
You can set different types of icons using the **MessageBoxIcon** enumeration. Here are the common options:

- **MessageBoxIcon.Information**: A blue information icon (i).
- **MessageBoxIcon.Warning**: A yellow triangle with an exclamation mark.
- **MessageBoxIcon.Error**: A red "X" icon indicating an error.
- **MessageBoxIcon.Question**: A question mark icon.

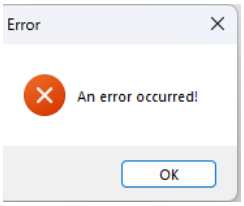
For example:

```
DialogResult result = MessageBox.Show("Do you want to continue?", "Confirmation", MessageBoxButtons.YesNo);
if (result == DialogResult.Yes)
{
    MessageBox.Show("You clicked Yes!", "Response");
}
else
{
    MessageBox.Show("You clicked No!", "Response");
}
```



For example:

```
MessageBox.Show("An error occurred!", "Error", MessageBoxButtons.OK, MessageBoxIcon.Error);
```



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Checking Rows Affected by Queries

The **cmd.ExecuteNonQuery()** method returns an integer indicating the number of rows affected by the executed query. If the query modifies one or more rows in the database, the returned value corresponds to the number of rows affected; otherwise, it returns 0.

```
int rowsAffected = cmd.ExecuteNonQuery();
```

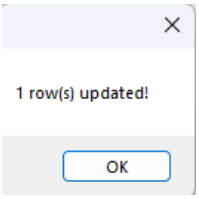
- rowsAffected will store the number of rows modified by the SQL command.
- If **no rows are affected**, it returns **0**.

For example:

```
string sql = "UPDATE department SET dname = 'Logistic' WHERE dno = 1";
using (MySqlCommand cmd = new MySqlCommand(sql, conn))
{
    int rowsUpdated = cmd.ExecuteNonQuery();
    MessageBox.Show(rowsUpdated + " row(s) updated!");
}
```

This displays a message box showing the number of rows that were updated. For example:

- If the UPDATE query affects 1 row, it will show a message like "1 row(s) updated!".
- If no rows are affected (i.e., no row has dno = 1), it will show "0 row(s) updated!".



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Parameterized Query

Parameterized Queries is the most effective method to prevent **SQL injection**. It separates the SQL code from the data, ensuring that user input is treated strictly as data, not executable code.

SQL injection is a type of attack that allows attackers to execute malicious SQL statements within a web application's query. These malicious statements can manipulate the database in dangerous ways, such as retrieving sensitive information, modifying data, or even deleting records.

For example: Deleting a record from the department table where the dno is 6.

1. **Using Placeholders in the SQL String**
   In the first line of code, a placeholder (also known as a parameter) is used instead of hard-coding a value.

   Syntax: The placeholder is identified by the @ symbol followed by a name (e.g., @dno).

```
string sql = "DELETE FROM department WHERE dno= @dno";
```

2. **Mapping Values with AddWithValue**
   The cmd0.Parameters.AddWithValue method is used to bridge the gap between the placeholder and the actual data variable.

   cmd0.Parameters.AddWithValue("@dno", dno); tells the application to replace the @dno placeholder in the SQL string with the value stored in the C# variable dno (which is 6).

```
int dno = 6;
cmd0.Parameters.AddWithValue("@dno", dno);
cmd0.ExecuteNonQuery();
```

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Adding Parameters SYNTAX

The **cmd.Parameters.AddWithValue** method is used to add parameters to a SqlCommand object when executing SQL queries.

SYNTAX

```
cmd.Parameters.AddWithValue("@parameterName", value);
```

Let's establish a naming convention for the controls.

Textbox:	**Txt**
List:	**Lst**
Button:	**Btn**
Label:	**Lbl**
Combo box:	**Cbo**
Image:	**Img**
Grid:	**Grd**

We will use the PascalCase naming convention where the first letter of each compound word in a variable is capitalized.